MILITARY SPECIFICATION

RDX/VINYL CHLORIDE COPOLYMER EXPLOSIVE COMPOSITION (PBX 9407) (FOR USE IN AMMUNITION)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions and packaging for the manufacture of two (2) types of high energy solid explosives composed of RDX, uniformly coated with a copolymer of trifluorochloroethylene and vinyl chloride (VC Copolymer) (see 6.1 and 6.5).

1.2 Classification.

1.2.1 Type designation. The following are the two types of explosives:

Type I shall consist of RDX coated with VC Copolymer

Type II shall consist of RDX coated with VC Copolymer plus calcium stearate.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves, Standard, for testing purposes

MILITARY

JAN-C-263 - Calcium Stearate
MIL-R-398 - RDX
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-650 - Explosive: Sampling Inspection and Testing
MIL-STD-1218 - ACS Chemicals.
MIL-STD-1235 - Single and Multilevel Continuous Inspection by Attributes.

US ARMY

7548644 - Box, Packing for High Explosives, Assembly, Details, Packing and Marking.

7548645 - Box, Fiberboard, Packing, Reusable, Collapsible for High Explosives

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation: E300 Recommended Practice for Sampling Industrial Chemicals.

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ASTM Designation: D2905 Number of Specimens Required to Determine the Average Quality.

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Constituent Materials. The constituent materials used in the production of Type I and Type II shall conform to the following:

RDX	- MIL-R-398, Type II, Class 5
Vinyl Chloride Copolymer	- See 6.5
Calcium Stearate	- Jan-C-263

3.2 Composition. For Type I, the explosive composition shall be RDX particles coated with the vinyl chloride copolymer. For Type II, the explosive composition shall be RDX particles coated with the vinyl chloride copolymer with calcium stearate added. Type I and Type II shall conform with the chemical and physical proportions given in Table I and 3.2.1 thru 3.2.3.

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TABLE 1. Requirements for RDX/PVC Explosive Compositions

	TYPE I		TYPE II		
	Batch	Lot	Batch	Lot	Test Paragraph
Composition					
RDX, %	94.0±1.0	94.0±0.5	93.5±1.0	93.5±0.5	4.6.1
VC Copolymer	6.0±1.0	6.0±0.5	6.0±1.0	6.0±0.5	4.6.2
Calcium Stearate, %			0.5±0.2	0.5±0.2	4.6.3
Volatiles, max, %	0.10	0.10	0.10	0.10	4.6.4
Bulk Density, min, g/cm^3	0.5	0.5	0.5	0.5	4.6.5

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3.2.1 Granulation. The explosive composition shall be in the form of free-flowing powder, which complies with the following granulation requirements when tested as specified in 4.6.6:

U. S. Standard Sieve No.	Cumulative Weight Percent Retained
18	0.5 max
30	10.0 max
60	90.0 min
80	98.0 min

3.2.2 Color. The color of the explosive composition shall be white to off-white at the time of acceptance when determined as specified in 4.6.7.

3.2.3 Foreign Matter. The explosive composition shall be free of metal inclusions, wooden particles and other foreign matter, when determined as specified in 4.6.8.

3.3 First article inspection. This specification contains provisions for first article inspection. When specified (see 6.2) the contractor shall furnish samples for first article inspection and approval. (See 4.3)

3.4 Workmanship. The manufacturers shall employ procedures which control and preclude conditions which may adversely affect the safety and reliability of the explosive (i.e. contaminated vessels, impurities, foreign matter, etc.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provision. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 5 kilograms of the explosive composition obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used during regular production.

4.3.2 Inspection to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.5 and 4.6 of this specification.

4.3.3 Rejection. See MIL-A-48078

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, a lot shall consist of one or more batches of one composition designation only, produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of explosive that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. In addition, each lot shall contain:

a. RDX of one type only, from one interfix lot number from one manufacturer only.
b. VC Copolymer from one interfix lot number from one manufacturer only.

4.4.2 Examination. Unless otherwise specified in the Classification of Defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II (See MIL-A-48078).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-R-63419 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548644 and 7548645
4.4.2.1	Box, Packing or Box, Fiberboard prior to closing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major					
131	Foreign matter		0.40%	3.2.3	Visual
132	Liner pierced or torn		0.40%	5.1	Visual
133	Liner improperly closed		0.40%	5.1	Visual
Minor					
201	Type of liner incorrect		0.65%	5.1	Visual
202	Bag liner pulled too tight not allowing slack		0.65%	5.1	Visual/Manual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548644
4.4.2.2	Sealed Box, Packing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major					
131	Top improperly assembled		0.40%	5.1	Visual/Manual
132	Box damaged		0.40%	5.1	Visual
133	Lot number misleading or unidentifiable		0.40%	5.1	Visual
134	Strapping missing, broken, or loose		0.40%	5.1	Visual/Manual
Minor					
201	Strapping improperly assembled		0.65%	5.1	Visual/Manual
202	Marking misleading or unidentifiable		0.65%	5.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-R-63419 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548645
4.4.2.3	Sealed Box, Fiberboard				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major					
131	Box torn or pierced		0.40%	5.1	Visual
132	Lot number misleading or unidentifiable		0.40%	5.1	Visual
Minor					
201	Marking misleading or unidentifiable		0.65%	5.1	Visual
202	Banding strips missing, broken or improperly applied		0.65%	5.1	Visual/Manual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.5 Testing.

PRECAUTION WARNING

This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Sampling for test 4.6.1 through 4.6.8. Approximately 500 grams of explosive composition shall be selected from each batch to be sampled using ASTM Procedures E300 for solids. Samples shall be selected for inspection in accordance with MIL-STD-1235, CSP-1 Plan, Inspection Level II, AQL 6.5%. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed shall be tested in accordance with the applicable methods given in paragraph 4.6. If any of these batches fail to meet any of the test requirements, that batch shall also be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted as required by MIL-STD-1235. The classification of defects shall be as given in Table II.

TABLE II
Classification of Defects

Inspection/Requirement	Defect Classification
Composition (see 3.2)	Major
Volatiles (see 3.2)	Major
Bulk Density (see 3.2)	Major
Granulation (see 3.2.1)	Major
Color (see 3.2.2)	Major
Foreign Matter (see 3.2.3)	Major

4.5.2 Inspection equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspecton equipment shall be employed, where applicable for all tests and examinations specified.

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The contractor is responsible for assuring proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes. (See 6.3)

4.6 Test methods and procedures. (See 6.4) The following tests shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks or ASTM D2905. Also, unless otherwise specified herein, all chemicals and reagents shall be ACS grade in accordance with MIL-STD-1218 or reagent grade.

4.6.1 RDX

4.6.1.1 Reagents - Toluene Solution.

Preparation of Saturated Toluene Solution. Add an excess quantity of RDX (about 5 g.) to 1 liter of Toluene and stir the slurry mixture for a minimum of 2 hours (room temperatue) with a magnetic stirrer. Store the Toluene slurry mixture over the excess RDX. Filter the slurry mixture prior to use.

4.6.1.2 Preparation of sample. Spread the sample in a thin layer on a suitable Petri dish (100 X 15 mm). Place sample on Petri dish in a vacuum oven at $60\pm2^{0}C$ for two (2) hours, or until constant weight is obtained.

4.6.1.3 Procedure. Weigh accurately (to nearest 0.1 mg), approximately a 10 g. portion of the dried sample and quantitatively transfer the sample into a 100 ml. Erlenmeyer flask. Add 50 ml. of the filtered Toluene solution to the flask. Stopper the flask and place the flask on a steam bath for 3 minutes and heat flask with contents. Remove flask from steam bath and place flask in a wrist-action shaker. Shake for 30 minutes. Quantitatively filter the contents of the flask through a clean, tared sintered glass crucible, medium porosity, 30 ml. Use five 15 ml. portions of the toluene solution to rinse the flask to assure quantitative transfer. Pour each 15 ml. portion of the toluene thru the crucible. Use suction for the filtering operation. Dry the crucible and contents for one (1) hour in a vacuum oven maintained at $70^{o}C\pm2^{o}C$. Remove crucible and contents from the oven and allow to cool in a desiccator. Reweigh the crucible and contents.

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4.6.1.4 Calculation. Calculate the percent RDX as follows:

$$\text{Percent RDX (and calcium stearate)} = \frac{(A-B)\ 100}{W}$$

Where: A = Weight of crucible and contents, g.
B = Tare weight of crucible, g.
W = Weight of sample, g.

4.6.2 VC Copolymer. Calculate the percent VC Copolymer by subtracting the value obtained in 4.6.1.4 from 100.0%.

4.6.3 Calcium Stearate. Quantitatively transfer an accurately weighed portion of the sample (approximately 5g. weighed to nearest 0.1 mg) to a 150 ml. beaker. Concurrently with this determination, conduct a blank determination without the sample but with all other materials. Add 30 ml. of concentrated hydrochloric acid (37-38%) to the beaker, cover with a watch glass, and heat beaker and contents on a steam bath for 15 minutes. Stir the contents of the beaker, occasionally, with a stirring rod while heating. After the 15 minutes, add an additional 30 ml. of concentrated hydrochloric acid, heat on steam bath and stir occasionally. While on steam bath, wash the cover glass and sides of the beaker with distilled water. Add, additional distilled water to the beaker to give a total of approximately 100 ml. solution. Remove watch glass and continue heating the solution on the steam bath for an additional 15 minutes. Quantitatively filter the solution in the beaker through a Whatman 40 paper into a 250 ml. volumetric flask. Wash the residue and beaker thoroughly with distilled water and add to the filtrate thru the filter paper previously used. Dilute the contents in the volumetric flask to the 250 ml. mark. Determine the calcium content (converted to calcium stearate) by atomic absorportion spectrophotometry using a suitable procedure given in a standard or reference chemical analysis textbook or handbook.

4.6.4 Volatiles. Determine the volatile content of the sample using method 101.6 of MIL-STD-650 at 60±2°C and a drying time of 2 hours.

4.6.5 Bulk density. Determine the bulk density of the sample using method 201.3 of MIL-STD-650.

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4.6.6 Granulation. Determine the granulation of the sample using method 204.1 of MIL-STD-650.

4.6.7 Color. Visually examine the color of the 500g. master sample.

4.6.8 Foreign matter. Visually examine the 500g. master sample to determine if foreign matter is present.

5. PACKAGING

5.1 Packing.

5.1.1 Level A. Packing shall be in accordance with Drawing 7548644 as specified for explosive composition.

5.1.2 Level B. (To be used for CONUS shipment only) Packing shall be in accordance with Drawing 7548645 as specified for explosive composition.

5.2 Marking. Marking shall be in accordance with Drawings 7548644 and 7548645 as applicable.

5.3 Special marking. The following marking shall be applied to the top and side (with other marking given in 5.2) of level B Packing "CONUS SHIPMENT ONLY".

6. NOTES

6.1 Intended use. The explosive composition covered by this specification is intended for use in the lead and booster of the Adam antipersonnel mine used in the M692/M731 HE 155mm projectile.

6.2 Ordering data.

a. Title, number and date of this detailed specification.
b. Quantity required and delivery schedules.
c. Type required (see 1.2)
d. First article requirements - Sample provisions (see 3.4 & 4.3)
e. Level of packing protection required.
f. Description sheets shall be prepared for each lot in accordance with MIL-STD-1171, DRSAR-P-702-109 or as directed in the contract.

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6.3 Submission of inspection equipment designs for approval. Submit equipment designs, as required to Commander, ARRADCOM, ATTN: DRDAR-QAR, Dover, New Jersey 07801.

6.4 Equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.5 Vinyl Chloride Copolymer. A material that has been found satisfactory for use in this explosive composition is FPC 461 made by Firestone Plastics Co. (formerly designated as Exon 461). FPC 461 is a copolymer of trifluorochloroethylyne and vinyl chloride. Use of VC Copolymer from another source should be evaluated by US Army Armament Research and Development Command, DRDAR-QAR-R before being approved as an equivalent.

Custodian:
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Preparing Activity:
Army-AR

Project Number: 1376-A166

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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